(5.1) $\forall \varphi.\ f\varphi = g\varphi \Rightarrow f = g.$

<u>Def 5.2</u>

a basis for $\text{epre}\,A$:= minimal set of states

$$\beta = \{|0\rangle, \dots, |n\rangle\}$$

(5.2) s.t. $\forall f, g$ process.

$$\forall i \in \beta.\ f i = g i \Rightarrow f = g$$

$$\dim(A) = |\beta| = n.$$

<u>Ex 5.4</u> $A = \{|0\rangle, \dots, |n\rangle\} \Rightarrow \beta_A = \{\{|0\rangle\}, \dots, \{|n\rangle\}\}$

<u>proof by contradiction</u>

$\exists j \in \{0, \dots, n\}.$

<u>Assume</u> $\beta_A - \{|j\rangle\}$ is a basis of A.

$\langle\{i,k\}|j\rangle = \langle\{i\}|j\rangle = 0$ but $\langle\{i,k\}| \neq \langle\{i\}|$

both processes of input epre A.

Def 5.5

(5.3) A set of state $\beta = \{|i\rangle\}_i$ is orthonormal

if $\delta_i^j = \langle j|i\rangle = \begin{cases} 1 & \text{if } i=j \\ 0 & \text{if } i \neq j \end{cases}$

if β forms a basis of type A,

then β is an orthonormal basis (ONB) of type A

Prop 5.10 orthonormal $\Rightarrow$ minimal

proof by contradiction

Theorem 5.14

Given all inputs, for all measurements,

the outcome (number) is the same,

(5.5) then the two processes are the same.

$\forall i \in \beta_{in}, \forall o \in \beta_{out}, \langle o|f|i\rangle = \langle o|g|i\rangle$

$\Rightarrow f = g$

(5.6) matrix of $f : A \to B$ basis basis

$$f := \left(f^j_i \,\middle|\, |i\rangle \in \beta,\ |j\rangle \in \beta' \right)$$

where $f^j_i := \langle j | f | i \rangle$ matrix entries

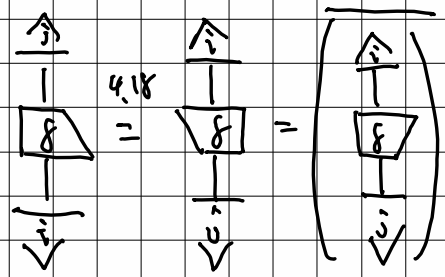
<u>Example 5.17</u> matrix representation of relation

$$R^j_i = \begin{cases} 1 & \text{possible} \quad (i,j) \in R \\ 0 & \text{impossible} \end{cases}$$

adjacency matrix $\Rightarrow$ connectivity

<u>Theorem 5.18</u> $\left(f^\dagger\right)^j_i = \overline{\left(f^i_j\right)}$

(4.18) $\langle \lambda \rangle = \langle \lambda \rangle^T$ number is self-transpose



Theorem 5.18 $\quad (f^\dagger)^j_i = \overline{(f^i_j)}$

adjoint = "conjugate-transpose"

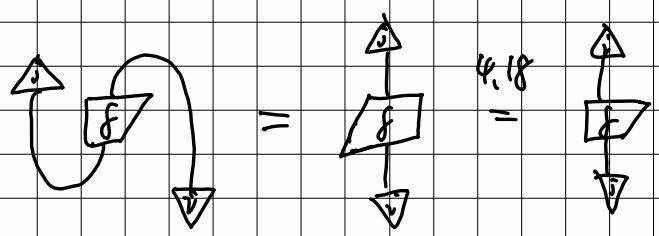
Theorem 5.19

① $(f^T)^j_i = f^i_j$

② $\overline{f}^j_i = \overline{(f^j_i)}$ $\quad \Rightarrow f^\dagger = \overline{(f^T)} = (\overline{f})^T$

where β and β' are self-conjugate

①

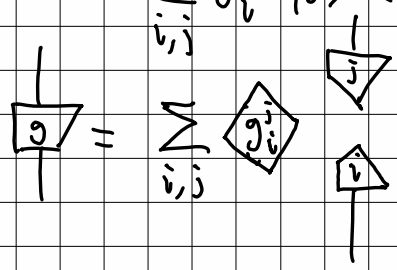


Exercise 5.20 $\quad$ self-adjoint process: $f = f^\dagger$

$$f^j_i = (f^\dagger)^j_i \overset{5.18}{=} \overline{(f^i_j)}$$

If some process theory allows "overlaying" processes as <u>sum</u> of processes, a matrix $g_{N \times M}$ can be expressed as

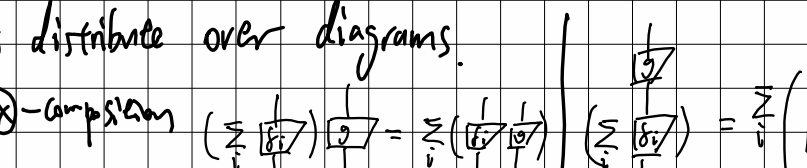
$$(5.9) \quad g = \sum_{i,j} g^j_i \, |j\rangle\langle i|$$



<u>Def 5.21</u>

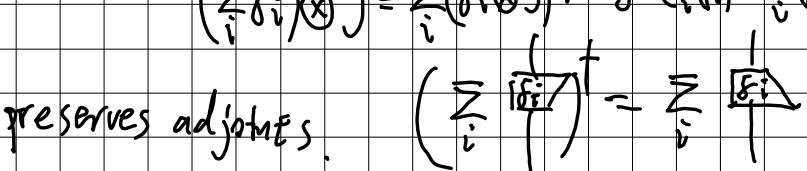
1. Sum of two processes
   1. unit 0 — $0 + f = f = f + 0$
   2. associative — $(f+g)+h = f+(g+h)$
   3. Commutative — $f + g = g + f$

2. Sums distribute over diagrams.
   1. $\otimes$-composition
   2. $\circ$-composition



$$\left(\sum_i f_i\right) \otimes g = \sum_i (f_i \otimes g) \qquad g \circ \left(\sum_i f_i\right) = \sum_i (g \circ f_i)$$

3. Sums preserves adjoints.



Theorem 5.22

$$\left(\sum_k f_k\right)_i^j := \sum_k (f_k)_i^j$$

$$\langle j | \sum_k f_k | i \rangle = \sum_k \langle j | f_k | i \rangle$$

Example 5.25 Sum for relation

is like logical disjunction ($\vee$) on Boolean.

$$0+0=0$$
$$0+1=1$$
$$1+0=1$$
$$1+1=1$$

while the $\circ$-composition for relation

is like logical conjunction ($\wedge$) on Boolean.

$(\circ)$ distributes over $(+)$: $x \circ (y+z) = x \circ y + x \circ z$

Def 5.28 Matrix Calculus of a process theory

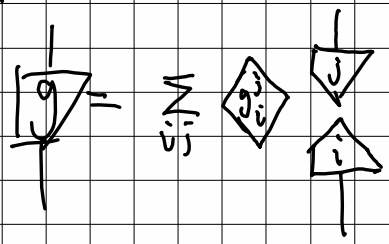
1. that admits string diagrams
2. has a (fixed, self-conjugate) ONB for every system type
3. has sums
   - unit law, associative, commutative
   - distribute over $(\circ)$ and $(\otimes)$
   - preserve adjoints

Matrices with operators

1. $(+)$, $(\circ)$, $(\otimes)$
2. $(\dagger)$, $(\overline{\ })$, $(\overline{\dagger})$

Theorem 5.29

A process g with fixed ONB B and B'

(5.14) 

has matrix $\underset{\text{output } B'}{N} \times \underset{\text{input } B}{M}$

$$g = \sum_{ij} g^j_i \, |j\rangle\langle i|$$

$$= \begin{pmatrix} g^1_1 & & g^1_m \\ & \ddots & \\ g^n_1 & & g^n_m \end{pmatrix}$$

~~proof~~ $\left(g_{N\times M}\right)^j_i = \langle j | g | i \rangle$

$$= \langle j | \left(\sum_{kl} g^l_k |l\rangle\langle k| \right) | i \rangle$$

$$= \sum_{kl} g^l_k \underbrace{\langle j | l \rangle}_{=\delta^j_l} \underbrace{\langle k | i \rangle}_{=\delta^k_i}$$

$$= \underset{(k=i \wedge l=j)}{g^j_i} + \underset{(k\neq i \vee l \neq j)}{0}$$

$$= g^j_i$$

<u>Theorem 5.30</u> $\mathcal{B} = \{|i\rangle\}_i$

ONB $\mathcal{B}$ for $A \iff \mathcal{B}$ spans A

(5.16) $\forall |\varphi\rangle \in A.\ |\varphi\rangle = \sum_i \lambda_i |i\rangle$